Physical Chemistry Cume

For the decomposition of silver carbonate in air at 1 bar

$$Ag_2CO_{3(s)} \rightleftharpoons Ag_2O_{(s)} + CO_{2(g)}$$

the following thermodynamic properties are available

	Molar entropy, S^0_m (298K) ($JK^{-1}mol^{-1}$)	Molar enthalpy of formation $\Delta_f H^0$ (298K) ($kJmol^{-1}$)	Molar heat capacity, $C_{p,m}$ ($JK^{-1}mol^{-1}$)	melting point (K)
$Ag_2CO_{3(s)}$	167.36	-501.7	109.6	483
$Ag_2O_{(s)}$	121.3	-31.1	65.9	553
$CO_{2(g)}$	213.8	-393.5	37.1	195 (sublimation)

(a) What is the change in entropy for the reaction at 298 K and 383 K.

(30 pts)

(b) If 1 g of $Ag_2CO_{3(s)}$ (Mol. wt. = 275 .75 $gmol^{-1}$) is present in a reaction vessel, what partial pressure of $CO_{2(g)}$ must be present in the air to ensure that the decomposition of $Ag_2CO_{3(s)}$ does not occur (i.e. is not spontaneous) at 383 K (110°C). Assume ΔH and C_p are independent of temperature.

(50 pts)

(c) In which direction will the equilibrium shift if 0.1 g of $Ag_2O_{(s)}$ (Mol. wt. = 231 .74 $gmol^{-1}$) is added to the reaction mixture from (b)?

(20 pts)